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DHANALAKSHMI SRINIVASAN UNIVERSITY

TIRUCHIRAPALLI – 621 112 TAMIL NADU, INDIA

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School of Engineering and Technology

Department of Electronics and Communication Engineering

CONTINUOUS ASSESSMENT TEST – I – August 2026

AY: 2026-27 ODD SEMESTER

Register No.:	**Year/Semester:** II/III&III/V
Course Code: 24OEC925	**Maximum Marks:** 50
Course Title: Introduction to Sensor Technology	**Time:** 90 MINS
Date & Session: 18-08-2026 FN/AN	**SET:** A

Course Outcomes:

COs	COURSE OUTCOME
CO1	Introduce students to the principle of various Transducers, their construction applications and principle of operation, standards and units of measurements
CO2	To study about Analog Signal Conditioning

Q.No.	PART – A Conceptual Questions (10×2=20 Marks)	Bloom's Level	Mapping COs	Marks
1	Define Sensors	BL1	CO1	2
2	Define Transducer.	BL1	CO1	2
3	Describe Linearity.	BL2	CO1	2
4	Explain why Hysteresis is Undesirable.	BL1	CO1	2
5	List out the types of errors in Instrumentation systems.	BL1	CO1	2
6	Name any two AC bridges used for measuring inductance or capacitance.	BL1	CO2	2
7	Differentiate between an attenuator and an amplifier.	BL2	CO2	2
8	State any two characteristics of an ideal operational amplifier.	BL1	CO2	2
9	Define the voltage gain of an inverting amplifier	BL2	CO2	2
10	Analyze the High Input Impedance of an Instrumentation Amplifier.	BL2	CO2	2

$1+\frac{2R_2}{R_{G}}$ $\frac{R_f}{R_1}$ CMRR

Q.No.	PART- B Application Questions (2×15=30 Marks)	Bloom's Level	Mapping COs	Marks
11a	Describe the static characteristics of transducers such as accuracy, precision, sensitivity, linearity, hysteresis, resolution, repeatability, and reproducibility with suitable illustrations.	BL2	CO1	15
	Or			
11b	Discuss the various types of measurement errors. Explain the causes, effects, and methods to minimize gross, systematic, and random errors with suitable examples.	BL2	CO1	15
12a	Evaluate the performance of bridge circuits used in electrical measurements. Compare the Wheatstone bridge, Maxwell Bridge, Schering Bridge, and Wien Bridge with suitable applications.	BL2	CO2	15
	Or			
	Differentiate between differential amplifiers and instrumentation amplifiers. Analyze their circuit configurations, performance characteristics, and applications.	BL2	CO2	15

Bloom's Level-wise Mark Distribution:

Remember	Understand	Apply	Analyze	Evaluate	Create	Total
12	23	15				50

Course Outcome wise Mark Distribution

COs	Remember	Understand	Apply	Analyze	Evaluate	Create	Total
CO1	10	15					25
CO2	2	8	15				25
Total	12	23	15				50

Prepared by

Approved by